

Spin-cooling of the motion of a trapped diamond

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Observing and controlling macroscopic quantum systems has long been a driving force in quantum physics research. In particular, strong coupling between individual quantum systems and mechanical oscillators is being actively studied^{1–3}. Whereas both read-out of mechanical motion using coherent control of spin systems^{4–9} and single-spin read-out using pristine oscillators have been demonstrated^{10,11}, temperature control of the motion of a macroscopic object using long-lived electronic spins has not been reported. Here we observe a spin-dependent torque and spin-cooling of the motion of a trapped microdiamond. Using a combination of microwave and laser excitation enables the spins of nitrogen–vacancy centres to act on the diamond orientation and to cool the diamond libration via a dynamical back-action. Furthermore, by driving the system in the nonlinear regime, we demonstrate bistability and self-sustained coherent oscillations stimulated by spin–mechanical coupling, which offers the prospect of spin-driven generation of non-classical states of motion. Such a levitating diamond—held in position by electric field gradients under vacuum—can operate as a ‘compass’ with controlled dissipation and has potential use in high-precision torque sensing^{12–14}, emulation of the spin-boson problem¹⁵ and probing of quantum phase transitions¹⁶. In the single-spin limit¹⁷ and using ultrapure nanoscale diamonds, it could allow quantum non-demolition read-out of the spin of nitrogen–vacancy centres at ambient conditions, deterministic entanglement between distant individual spins¹⁸ and matter-wave interferometry^{16,19,20}.

Since the experiment by Einstein and de Haas in 1915²¹, much work has been carried out on the detection of atomic spins through mechanical motion²², culminating in the observation of a magnetic force from single spins^{10,11} and magnetometry at the nanoscale¹³. Conversely, single spins and qubits have also been used to sense the motion of objects. Single-qubit thermometry of mechanical oscillators was realized using a superconducting qubit coupled to membranes^{4,8} and nitrogen–vacancy (NV) centres coupled to cantilevers^{5–7}. A crucial next step is to reach strong coupling between long-lived spins and mechanical oscillators, which will enable ground-state cooling, as in tethered quantum opto-mechanical platforms^{23–25}, and the observation of quantum superpositions of macroscopic systems². One further prospect is the entanglement between multiple spins¹⁸, with far-reaching implications for quantum information science and metrology²⁶. Obtaining coupling rates that surpass the decoherence of both the spin and the mechanical system is however still a challenge for most state-of-the-art platforms. Recently, there has been renewed focus on levitating objects^{27,28} motivated by the low mass and high *Q*-factors that they offer, together with the possibility of cooling their motion using embedded spins. There is a strong analogy between this platform—where spins move a levitating crystal—and laser-cooled atoms, where electrons move atomic nuclei. It may thus be forecast that a levitating particle

containing a few long-lived spins will ultimately reach a level of control similar to that of trapped ions³, with bright prospects for the above-mentioned applications.

In this work, we report a controllable torque induced by the spins of atoms embedded in a microscale object. Specifically, we couple the spin of many NV centres to the orientation of a trapped diamond particle. This coupling then enables us to show mechanical read-out of the spin resonance of the NV centres together with cooling and lasing of the diamond motion.

The crystallographic structure of the NV centres is depicted in Fig. 1A. The spin–spin interaction between the two electrons in the NV-centre ground state lifts the degeneracy of the spin-triplet eigenstates by $D = 2.87$ GHz at room temperature. Such an interaction implies that the NV centre has a preferential quantization axis that is along one of the four crystal axes $\langle 111 \rangle$. In the presence of a magnetic field $\mathbf{B}$ at an angle ϕ with respect to the NV axis, the energy difference between the two energy eigenstates $|m'_s = \pm 1\rangle$ is about $\gamma_e B \phi$, where γ_e is the gyro-magnetic ratio of the electron. Spin control can then be performed using optical and microwave excitation, and the angular dependence of the NV spin energy eigenstates is expected to allow rotation and cooling of the diamond angular motion. Once in a magnetic state via a resonant microwave excitation, the NV centre will tend to align the

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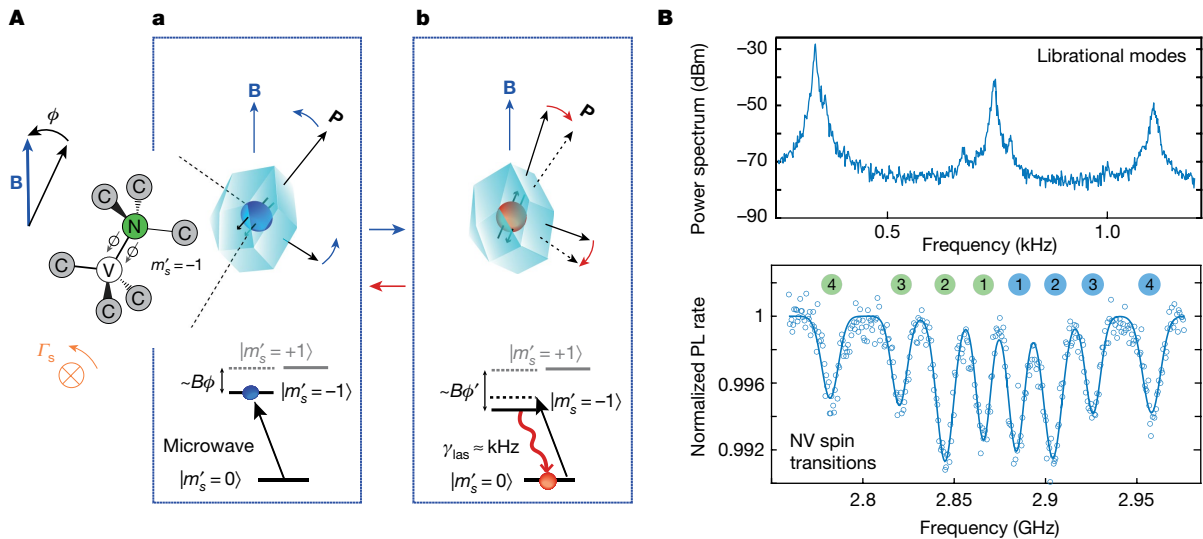


Fig. 1 | Spin and mechanical systems. **A**, Sketch of the diamond crystallographic structure hosting NV defects and the effects of rotation. **a**, Equilibrium position of the diamond in the Paul trap before the microwave excitation. The principal diamond axis **P** points towards the main trap axis. The ground state spin levels and microwave drive are depicted below. The magnetic field **B** lifts the degeneracy between the excited spin states by about $\gamma_e B \phi$, where ϕ is the angle between the magnetic field and the NV axis. The microwave signal prepares the NV in a magnetic state, which induces a torque on the diamond. Inset: left, definition of ϕ ; right, representation of the NV (N, nitrogen; V, vacancy; C, carbon). **b**, Once at the new angular position, the spin projection onto the magnetic field is changed to $\gamma_e B \phi'$. The microwave

corresponding diamond crystalline axis to the magnetic field, as illustrated in Fig. 1A, a. Further, laser-triggered relaxation from the excited state can then extract the work exchanged between the spin magnetic energy and the librational motion (see Fig. 1A, b).

In our experiment, harmonic librational (sometimes called torsional, pendular or rotational) confinement is provided both by the Paul trap (Methods) and the particle asymmetry. We measure the libration of the diamond by using the reflection of the laser from the diamond surface. The micrometre-size roughness of our 15- μm particles enables a specular pattern to be detected at the particle image plane, which after mode-matching one of the many bright spots to an optical fibre yields an angular sensitivity of about $0.3 \text{ mrad Hz}^{-1/2}$ and a resolution of about $10 \text{ mrad Mcounts s}^{-1}$ (see Methods and Extended Data Fig. 4). Under vacuum conditions ($\sim 1 \text{ mbar}$), the signal power spectrum plotted in Fig. 1B shows harmonic motion of the three librational modes with frequencies $\omega_{\phi}/2\pi$ ranging from 200 Hz to 1.2 kHz and with a damping rate of about 15 Hz. Figure 1B also shows an ODMR spectrum for a diamond outside the trap, in the presence of a magnetic field $B \approx 30 \text{ G}$. Eight transitions, corresponding to the projections of the **B** field onto the four NV orientations, are observed, with typical spin decoherence rates $1/T_2^* \approx 7 \text{ MHz}$.

We now measure the diamond rotation induced by the $N \approx 10^9$ NV electronic spins inside the diamond, with the same optical read-out as for the librational mode detection, as depicted in Fig. 2A, a. The expected magnitude of the spin torque is $\Gamma_s = \hbar N \gamma_e B S_z \approx 10^{-19} \text{ N m}$. Here S_z is the population in one of the magnetic states, determined by the competition between the microwave and laser polarization (both at rates in the 100 kHz range, see Methods and Extended Data Fig. 5). This torque gives a displacement of the particle angle in the trap, $\delta\phi = \Gamma_s / I \omega_{\phi}^2 \approx 10 \text{ mrad}$, where $I \approx 10^{-22} \text{ kg m}^2$ is the particle moment of inertia. As can be seen in Fig. 2A, b, sweeping a microwave frequency around the spin resonances indeed enables conspicuous features to appear. Once in the magnetic state $|m'_s = -1\rangle$ or $|m'_s = +1\rangle$, the NV centres tend to align or anti-align the diamond orientation to the magnetic field, which is manifest in the anti-correlation between the detected

frequency is then no longer resonant: the spin relaxes to the ground state and the diamond returns to its initial position. The red wavy arrow represents the polarization at a rate γ_{las} from the excited state to the ground state induced by the green laser. **B**, Measurements of the three librational modes undergoing Brownian motion at 1 mbar of vacuum pressure (top) and of the typical electron spin resonances from the NV ensemble within a microdiamond outside the trap using standard optically detected magnetic resonance (ODMR) at 30 G (bottom; PL, photoluminescence). The filled circles label the four orientations of the NV spins. Green (blue) circles correspond to the $m_s = 0$ to $m_s = -1$ ($m_s = +1$) transitions. Solid lines are a fit to the data.

intensity levels for all pairs of transitions. A standard ODMR also measured under the same magnetic field amplitude and measurement time (see Fig. 2A, c) demonstrates perfect correlation of the frequencies of the peaks in the two measurements.

This spin-mechanical effect is in fact much richer than a static spin-dependent torque. As shown in Fig. 1A, the NV centres are magnetized through a microwave tone whose detuning from the NV resonances changes as the diamond rotates. To first order, such a torque will increase (decrease) the confinement of the Paul trap if the microwave frequency is blue (red) detuned from the spin resonance at the equilibrium angular position. Further, since the spin lifetime is of the order of the libration period a delay between the NV magnetization and the angular oscillation, observed in ref. ²⁹, can indeed induce a torque that depends on the velocity, in close analogy with opto-mechanical schemes^{23–25} and with Sisyphus cooling of cold atoms. The net result is a pronounced cooling (heating) of the diamond motion when the microwave is red (blue) detuned from the spin resonance as sketched in Fig. 2B. In order to observe such spin-spring and spin-cooling effects, we monitor the librational power spectrum as a function of the microwave detuning from the electronic spin resonance. Figure 2B also shows the result of measurements taken for three different microwave frequencies. A strongly modified spring and damping of the mechanical mode are observed. Assuming that the initial temperature is 300 K (see Methods), the resulting temperature after spin-cooling is here 80 K. Figure 2C shows measurements of the damping rate and spring effects as a function of microwave frequency in good agreement with a theoretical model (see Methods and Extended Data Fig. 1). Cooling is ultimately limited by heating from the microwave excitation of the motion on the blue side. This could be eliminated by increasing the trapping frequency $\omega_{\phi}/2\pi$ above the NV spin-transition linewidth.

We now make a step into a regime where the spin-mechanical interaction induces nonlinear effects on the librational mode. With a stronger spin torque (see Methods), Fig. 3A, a displays the expected bistable behaviour for the angular degree of freedom when the microwave

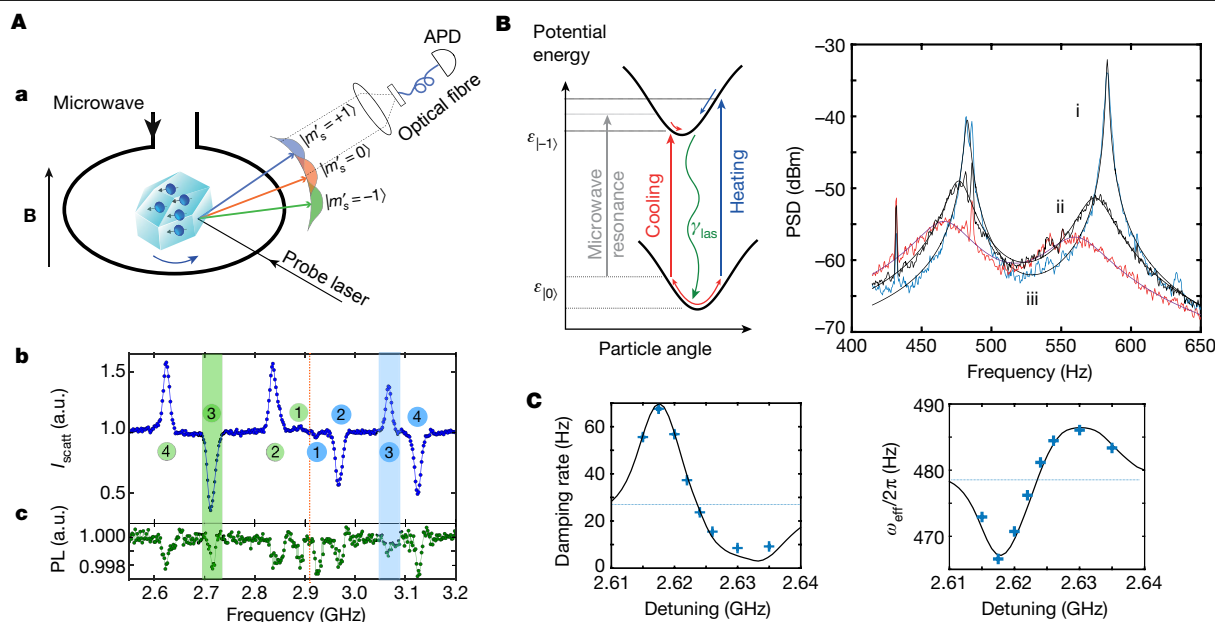


Fig. 2 | Spin-dependent torque and cooling of a levitating diamond.

A, Rotation of a diamond particle when NV centres are in a magnetic state. **a**, Sketch of the laser beam deflection induced by the NV spin torque. APD, avalanche photodiode. **b**, Detected APD count-rate I_{scatt} as a function of the microwave frequency. The green (blue) filled circles correspond to the $m_s = 0$ to $m_s = -1$ ($m_s = +1$) transitions, while the green (blue) bars highlight the NV orientations that rotate the diamond as per sketch **a**. **c**, Corresponding ODMR. a.u., arbitrary units. **B**, Left, cooling/heating cycle of the librational motion induced by the spin-mechanical coupling. Right, power spectrum of the

detected light intensity reflected from the diamond surface when the microwave frequency is tuned to the blue (trace i), to the centre (trace ii), and to the red (trace iii) of the spin resonance. The alignment of the reflected light from the diamond surface in the fibre was optimized to only let these two librational modes appear in the power spectrum. Note that the particle is here different from the one used in Fig. 1B. PSD, power spectral density. **C**, Effective damping rates (left) and librational mode frequencies $\omega_{\text{eff}}/2\pi$ (right) as a function of the microwave detuning. Lines show a fit to the experimental data using numerical simulations.

frequency is scanned across the spin resonance. The angle can be found at two metastable positions A or B depending on the history of the angular trajectory (see Supplementary Information). The hysteresis behaviour is indeed observed in the experiment, and shown in Fig. 3A, b.

The evolution of the particle orientation over time at a fixed microwave tone is also plotted in Fig. 3A, c. We note that the particle orientation jumps from site A to B in a seemingly unpredictable manner owing to random kicks given to the particle. The average population at the

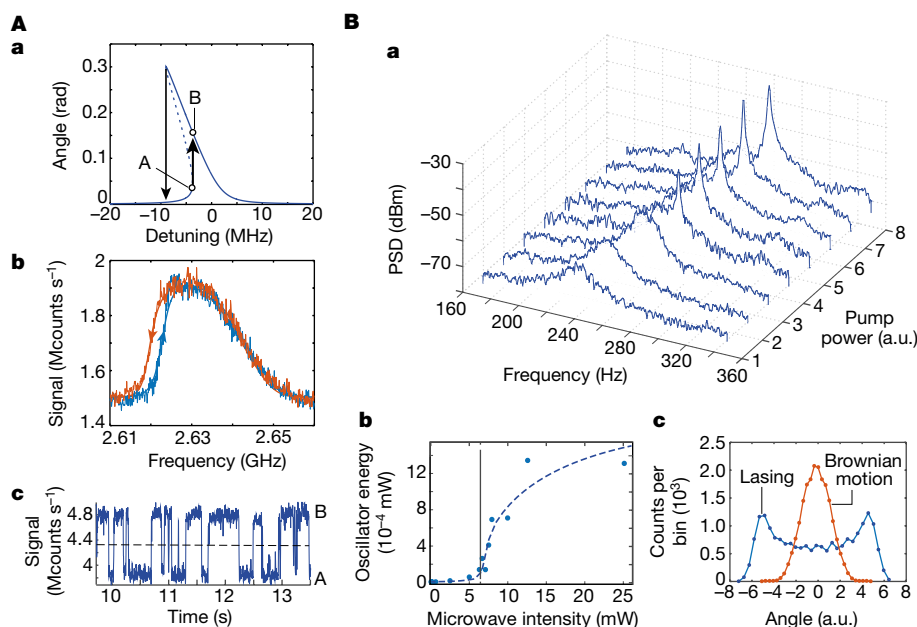


Fig. 3 | Nonlinear spin-motion dynamics. **A**, Bistability. **a**, Evolution of the particle angle as a function of the microwave detuning (see below for points A and B). **b**, Hysteresis behaviour of the particle orientation when the microwave signal is scanned from the red to the blue (blue curve) or from the blue to the red (red curve), as indicated by the arrows. **c**, Particle orientation as a function of time for a fixed microwave frequency tuned to the hysteretic frequencies

(2.625 GHz), showing angular jumps between two stable sites A and B. **B**, Phonon lasing. **a**, Evolution towards lasing of the power spectrum of the librational motion as a function of the microwave power. **b**, Oscillator energy as a function of microwave intensity using microwave powers ranging from -44 dBm to -16 dBm in steps of 4 dBm. The dashed line is a fit to the data. **c**, Histogram of the Brownian and lasing angular motions.

angular position B can also be studied as a function of microwave detuning, and was shown to increase as the microwave frequency is tuned towards the blue side of the spin resonance (see Extended Data Fig. 6 and Supplementary Information).

We now set the microwave frequency to the blue side of the spin resonance in this strong spin-torque regime. Figure 3B, a shows the power spectral density as a function of the microwave pump power, where a transition from Brownian motion to a self-sustained oscillation is observed (see also Extended Data Fig. 3). Such a lasing-like action of a mechanical oscillator was observed in the first radiation pressure cooling experiments⁶ with proposed applications in metrology. The spin-mechanical gain that enables such lasing action here is provided by blue-detuned microwave excitation, which amplifies the angular motion up to a point where losses are compensated by the magnetic gain (see Extended Data Fig. 2 for numerical results). The oscillator energy as a function of the microwave power is shown in Fig. 3B, b. A lasing threshold is observed at 6 mW of microwave excitation. Another signature of mechanical lasing is shown in Fig. 3B, c, which displays the probability distribution of the angular degree of freedom with and without microwaves. Under blue detuned microwave excitation, the probability distribution departs from the Gaussian process (red curve) for Brownian motion, and turns into the characteristic probability distribution of a coherent oscillation (blue curve). This effect shows that the librational mode can operate stably, deep in the nonlinear regime, and highlights further the analogy between the present spin-mechanical platform and opto-mechanical systems.

Coupling individual spins to the motion of a macroscopic oscillator will have far-reaching applications in fundamental science, quantum information and metrology. The present spin-dependent torque itself may be used for detecting atomic defects with electron spins that cannot be efficiently detected through ODMR. Further, the approach may also be applied to other torsional nano-mechanical platforms³⁰, which can exploit the long NV spin–lattice relaxation at low temperatures for longer interrogation times and efficient cooling. Last, operation in the resolved sideband regime where $\omega_p/2\pi \gg 1/T_2^*$ could be realized after modest improvements to the present set-up. We estimate that using a 1- μm -diameter pure diamond grown by chemical vapour deposition (CVD) attached to a 1- μm -diameter ferromagnet would enable the resolved sideband regime to be entered for this hybrid structure. Librational frequencies $\omega_p/2\pi$ above 200 kHz have indeed been observed recently²⁹ and NV centres with $1/T_2^* = 50$ kHz electron-spin decoherence rates can readily be obtained in CVD-grown microdiamonds enriched in ^{12}C . Entering this regime would offer the immediate prospects of ground-state cooling the diamond libration and multipartite spin entanglement, and would provide strong impetus to bridge the gap between trapped particles and trapped atoms.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-020-2133-z>.

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Methods

Microdiamond properties

The diamonds that we use are in the form of a powder, with particles that have a diameter of 15 μm . They are supplied by the company Adamas, which produces diamonds with a concentration of NV centres in the range 3–4 p.p.m., corresponding to $(1.5\text{--}2) \times 10^9$ NV centres per microdiamond. Using the same collection optics as was used in ref. ³¹, we estimated the number of optically addressed NV centres to lie in that same range. This is 4 to 5 orders of magnitude larger than the concentration used in the experiments reported in refs. ^{31–33}, where no spin torque was observed. Under continuous laser excitation at around 10 μW , our diamonds started to heat up at 0.1 mbar, which is similar to pressures used in³²; this points to the role played by impurities other than NV centres in the heating observed in ref. ³².

The Paul trap

We operate with a Paul trap that is similar to the one used in ref. ³¹ except that the particles are stably trapped at the bottleneck region of the trap, where both the electric field gradient and the anisotropy are stronger, yielding higher librational confinement. The pressure we operate at is 1 bar for the spin-torque measurements shown in Fig. 2, and in the millibar range for the cooling and phonon lasing experiments. Below 1 mbar, the diamond starts to rotate owing to a locking mechanism induced by the Paul trap drive, making it impossible to observe the spin-dependent torque, which relies on very stable libration.

NV spin polarization and read-out

Owing to an intersystem crossing in the excited state of the NV centres, the electronic ground state $|0\rangle$ is brighter than the $|\pm 1\rangle$ states under green laser illumination. This provides a means to read out the Zeeman splitting by scanning a microwave tone around the resonance, carrying out ODMR. Here, the microwave is applied directly to the trapping electrode, which provides an efficient means to excite the spins. The photoluminescence is detected using standard confocal microscopy. We use about 100 μW of laser light at 532 nm to polarize the NV centres. The laser is focused via a lens inside the vacuum chamber which has a numerical aperture of 0.5 and a working distance of 8 mm. The focal point of the laser is kept a few tens of micrometres away from the microdiamond to mitigate the effect of radiation pressure and to enable laser excitation of the whole diamond^{31,33}. To measure the polarization rates to the ground and to the excited magnetic states, we carry out the sequence depicted in Extended Data Fig. 5a. The photoluminescence rate is measured as a function of the time τ for both sequences and is plotted in Extended Data Fig. 5b. The laser induced polarization rate to the $m'_s = 0$ state is 3.3 kHz. The microwave polarization rate $\Gamma_M = \Omega^2 T_2^*$ (see Supplementary Information) to the magnetic state $m'_s = -1$ is found to be 8 kHz when using -5 dBm of microwave power measured before a 25 dB amplifying stage. An estimate based on both the ODMR width and a Ramsey sequence yields $T_2^* = 70$ ns, implying $\Omega/2\pi = 60$ kHz.

The degree of spin polarization cannot be estimated precisely without using a full numerical model and the 8 rate equations including mixing by the magnetic field transverse component. The magnetic field transverse component reduces the polarization time owing to mixing of electron spin states both in the ground and excited levels. This enhances the probability of non-radiative crossing to the metastable level and reduces the ODMR contrast by³⁴ 30%. Overall, this reduces the degree of optical polarization to the $m'_s = 0$ spin state to about 60%.

Detection and analysis of the libration

The diamond motion is detected by collecting the back-reflected green light from the diamond surface, separated from the excitation light using a polarizing beamsplitter. The best sensitivity is achieved by taking advantage of the speckle pattern produced by the rough surface of

the microdiamond under coherent illumination. At the particle image plane, which is located a few tens of centimetres away from the particle, an image is formed with an additional speckle feature. To detect the diamond motion, we focus a small area of this image onto a single-mode optical fibre and detect the photons transmitted through the fibre with a single-photon avalanche photodiode. The detected signal is then highly sensitive to the particle position and orientation.

Angular displacement sensitivity

For a given levitating particle, we can optimize, in real time, the change in the optical signal coming from the angular displacement of the particle by selecting the most favourable region of the particle image. To do this, we look at our optical signal while switching a microwave field tuned to one ODMR transition at a frequency of 1 Hz. Alignment is done by maximizing the change in the coupled light intensity as the diamond jumps between two angular positions. The linearity of the coupled light with the rotating angles can finally be assessed by looking at a higher order of the harmonic motion once the libration frequencies are identified.

While being a sensitive measurement of the angular displacement, our technique does not give an absolute measurement of the angle change. The spin-torque vector Γ_s is orthogonal to the plane defined by the magnetic field and the NV axis (it tends to align the NV axis to the **B** field). However, because the angular confinement is not isotropic, the particle rotation axis is not necessarily collinear with the spin torque. Determining the exact three-dimensional rotational dynamics of the particle would necessitate knowledge of the orientation of the NV axes with respect to the principal axes of the angular motion.

Using NV magnetometry, the mechanically detected spin resonance can nonetheless be used to relate the optical signal change to the angular displacement of the particle. A set of three mechanically detected spin resonances corresponding to three different microwave powers are shown in Extended Data Fig. 4a, under a magnetic field of 144 G. The minimum of each curve falls on the dashed line. The lower panel of Extended Data Fig. 4a is a theoretical curve where the angle between the NV axis and the magnetic field direction is plotted as a function of the frequency of the NV spin transition. This curve is obtained by diagonalizing the NV spin Hamiltonian in the presence of a magnetic field of 144 G. Since the maximum magnetization of the NV spins is obtained when the microwave field is resonant with the spin transition, one can relate the maximal change in the optical signal (ΔS) to the variation of the angle between the NV axis and the magnetic field direction ($\Delta\theta_{\text{NV}}$). Doing so, we obtain here a resolution of 43 mrad Mcounts⁻¹s⁻¹. Extended Data Fig. 4b shows a time trace of the optical signal upon Brownian motion of the particle. From the standard deviation of this signal and the above calibration, we obtain an angular displacement sensitivity of 0.3 mrad Hz^{-1/2}.

These numbers are however only upper bounds for our resolution and sensitivity. To explain why this is the case, Extended Data Fig. 4c shows a sketch of the angular motion of the diamond after magnetizing one class of NV spins. For simplicity, we consider rotation about two axes here. In a reference frame with axes given by the principal librational mode directions, we can parametrize the orientation of the NV axis in a subspace defined by the two angular coordinates θ_x and θ_y . The orientation without magnetization ($M_z = 0$) is given by the trap and particle geometry and labelled O. The point B in this space is the direction of the magnetic field. Upon magnetization, a torque is applied to the particle such that the orientation follows the OB trajectory over time. However, owing to different confinement of the librational modes ω_x and ω_y along the *x* and *y* axes, the angular motion takes place along a different trajectory.

In our experiments, the orientation of the magnetic field and NV axes relative to the principal axis of the libration is unknown. This prevents us from fully calibrating our detected angular motion. Nevertheless, provided that the detection is optimized to the librational mode having

the highest confinement, we can ensure that the detected angular displacement θ_d is smaller than the angular displacement θ_{NV} sensed by the NV spins. This can be seen in Extended Data Fig. 4c, where we note “NV”, the equilibrium position when $\omega_x > \omega_y$. Our calibration method thus gives an upper bound to the obtained resolution and sensitivity. Optimization of the detection is performed by monitoring the power spectrum and tuning the speckle angle at the entrance of the fibre to maximize the power spectrum of the mode with the largest frequency. Extended Data Fig. 4d shows the power spectrum of the Brownian motion for two detection alignments. In the red trace, all three librational modes, indicated with black arrows, are clearly visible. In the blue trace, the detection is tuned to be mainly sensitive to the mode with the highest confinement frequency. The latter detection tuning is used for the data shown in Extended Data Fig 4a and b.

The sensitivity could be improved by collecting all the speckle pattern using a camera rather than just a fraction of it as we do now. Using a shorter laser wavelength would also straightforwardly improve the sensitivity. Another technical limitation comes from the trapped diamond motion in other modes than the libration mode of interest which adds noise to the angular displacement signal. In this regard, active stabilization of the centre of mass will greatly increase the sensitivity.

Power spectral density

Using the above described detection method, motional frequencies can be observed by sending the detected signal to a spectrum analyser. Under vacuum (1 mbar), the power spectrum exhibits narrow peaks at the trapping frequencies of the motional modes which are driven by Brownian motion (see Fig. 1B in the main text). For each librational mode, the power spectrum is fitted by the formula obtained in Supplementary Information:

$$S_\phi(\omega) = \frac{2\gamma kT}{I((\omega_\phi^2 - \omega^2)^2 + \gamma^2 \omega^2)} \quad (1)$$

The librational modes can in fact be unambiguously identified (and separated from the centre of mass modes) using the torque induced by the NV centres. By switching on and off a microwave field tuned to one spin resonance at the same period as that of one diamond libration, one performs parametric excitation of that librational mode. In our experiments, a sequence of five microwave pulses is enough to displace the angle above the Brownian thermal noise. Following a parametric excitation sequence, the diamond orientation ‘ring-down’, or decay, is observed. A typical decay curve is shown in Extended Data Fig. 1a. We typically find librational frequencies in the range 100 Hz to 1 kHz.

Parameters used for the spin-dependent torque measurements

The ODMR and spin-mechanical measurement scans shown in Fig. 2A of the main text are taken under atmospheric pressure. The green laser power was 330 μ W and the microwave power was set to 0 dBm. The magnetic field is around 95 G. For the mechanically detected spin resonance (Fig. 2A, b), the microwave detuning is scanned in 2 MHz steps with a duration of 10 ms per point. During those 10 ms, the diamond orientation has enough time to reach its equilibrium position and the spin-torque effect can be observed. The average count rate is $2.3 \times 10^6 \text{ s}^{-1}$ for a total averaging time of 10 min. For the ODMR trace (Fig. 2A, c), the microwave detuning is scanned in steps of 2 MHz with a duration of 1 ms per point. For each point, the microwave field is switched off for the first 0.5 ms and switched on for the last 0.5 ms during which the signal is acquired. This prevents mechanical effects from altering the detected photoluminescence signal from the NV centres. The photoluminescence count rate is $5 \times 10^6 \text{ s}^{-1}$ and the total averaging time for this measurement was 3 h.

Estimation of the temperature

The temperature associated with the librational modes can only be estimated. Obtaining a precise temperature value would require

knowledge about the moment of inertia of the particle, which is prone to strong systematic errors. The standard method is to vary the pressure³⁵ while observing the power spectrum: over the pressure range where its area is constant (to satisfy Liouville’s theorem under adiabatic transformation), the librational mode temperature is known to be 300 K as it is thermalized with the gas temperature. In our case, pressure variations slightly change the orientation and position of the trapped particle and, incidentally, the sensitivity to angular motion. This prevents such a method from being used. However two observations support the fact that the external degrees of freedom of the particle are thermalized at 300 K when operating in the millibar pressure range. We measured the particle internal temperature with our typical laser powers via NV thermometry³², and found it to be close to 300 K. This ensures that no heating of the libration modes comes from the heating of the gas surrounding the particle³⁶. We observed that heating of the particle by the laser starts below 0.1 mbar, similar to what was measured using diamonds that were doped with a three orders of magnitude smaller NV concentration. Several sources of noise could also heat up the particle, such as the laser-induced torque³³ or charge fluctuations. Heating by the former can be excluded as no noticeable changes of the power spectrum shape occur when laser power is increased up to 1 mW.

Parameters and calibration

The power spectrum of the detected librational modes depends strongly on the particle angle. For the same motional amplitude, a change in the particle angle potentially implies a different speckle pattern, which in turn changes the power spectrum sensitivity. Since the particle angle changes with the microwave detuning and power, different power spectra cannot be directly compared when the parameters are changed. Traces i, ii and iii in the right panel of Fig. 2B in the main text have thus been obtained at the same particle angle to enable quantitative comparison between their areas. Operating at the same particle angle was ensured by performing a resonant spin-mechanical detection at different microwave amplitudes and choosing pairs of microwave frequencies and powers that correspond to the same count rates. As shown in the data in Extended Data Fig. 1b, we chose microwave detunings corresponding to the points 1, 2, 3 for the two traces i and ii taken at microwave powers of –20 dBm and –10 dBm, respectively. The frequencies, which are 2.617 GHz, 2.623 GHz and 2.634 GHz, respectively on the blue, resonant and red side of the spin resonance all correspond to the same angle under these power conditions. A fit to the experimental curves in Fig. 2B was obtained using the formula

$$S_\phi(\omega) = \frac{2\gamma_{\text{eff}} kT}{I((\omega_{\text{eff}}^2 - \omega^2)^2 + \gamma_{\text{eff}}^2 \omega^2)} \quad (2)$$

The dependence of the damping γ_{eff} and frequency shift ω_{eff} on the microwave detuning shown in Fig. 2C was obtained using parametric excitation of the librational mode at 480 Hz. The microwave power is –10 dBm and for this measurement, the above-mentioned calibration issue (change in the sensitivity when the microwave detuning is varied) is not relevant. In order to extract the damping and shifts, the resulting ring-down was fitted by the formula:

$$S(t) = A_1 \sin(\omega_{\text{eff}} t + \phi) \exp(-\gamma_{\text{eff}} t/2) + A_2 \sin(\omega_2 t + \phi_2) \exp(-\gamma_2 t/2) + A_0 \quad (3)$$

where the second exponentially damped sine term takes into account the slightly excited librational mode at 590 Hz. Three of these ring-down traces are shown in Extended Data Fig. 1c. For each curve, the averaging time is around 100 s. An estimation of the temperature relies on comparing the damping with and without spin-cooling, using the relation²³ $T_{\text{eff}} = T - \frac{\gamma}{\gamma_{\text{eff}}}$, valid for small spin-spring frequency shifts.

Modelling of the experiment

For most of this paper, we modelled the experiment numerically using Monte Carlo simulations that include the full three-level structure of the NV spin-1 system in the ground state (see Supplementary Information). For the spin-cooling and spin-spring effects shown in Fig. 2C, the number of NV centres, polarization rate, Rabi frequencies, and angle between the NV centres and the main axis of the diamond are left as free parameters.

Bistability and phonon-lasing

The curve in Fig. 3A, a, shows the angular evolution of the particle as a function of the microwave detuning obtained using similar parameters to those in the linear regime, but using microwave and laser powers that best fitted the data in Fig. 3A, b, and a lower trapping frequency ($\omega_\phi/2\pi = 240$ Hz). Figure 3A, c, shows the evolution of the librational mode angle as a function of time in this regime. Several such curves were obtained for different microwave detunings, and are shown in Extended Data Fig. 6. A Monte Carlo simulation was also performed using our experimental parameters, with a microwave tuned to the red side, and shows similar jumps between the two stable points A and B. The data shown in Fig. 3B, a, of the main text show the evolution of power spectra for different microwave signal powers, when the microwave frequency is tuned to the blue of the ODMR transition. The onset of instability is seen at approximately 0 dBm. For a quantitative estimate of the threshold, we compute the area below the librational peak as a function of microwave detuning. This is shown in Fig. 3B, b. Note that here the sensitivity of the power spectra to the angle may induce some systematic errors. For these measurements, we fitted the data by the numerical model, and found good agreement with the numerical analysis, but a quantitative comparison with the experiment is difficult owing to the above mentioned angle-dependent sensitivity.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Author contributions T.D. and P.H. contributed equally to this work. T.D., P.H., L.N. and G.H. performed the spin-torque experiments; T.D., P.H. and G.H. analysed the data and performed the modelling with assistance from L.N., and G.H., T.D. and P.H. wrote the manuscript. All authors contributed to the interpretation of the data and commented on the manuscript.

Competing interests The authors declare no competing interests.

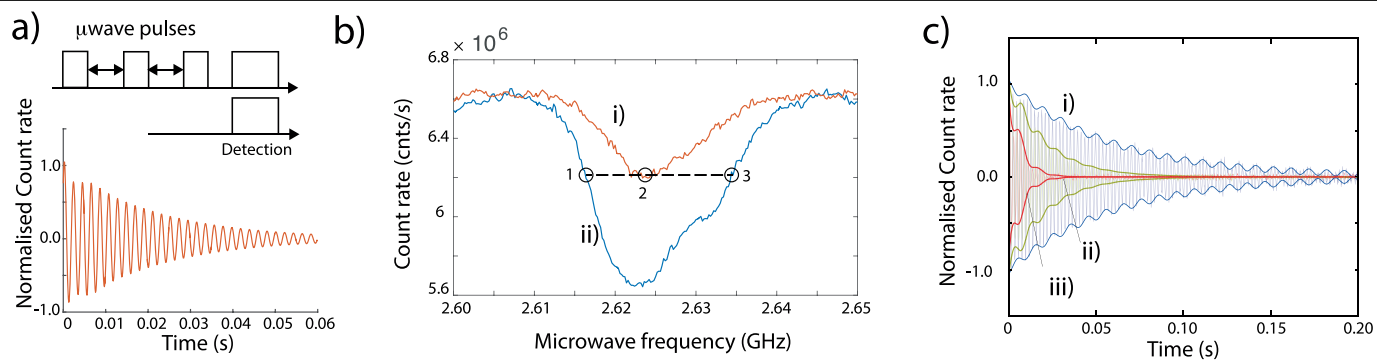
Additional information

Supplementary information is available for this paper at <https://doi.org/10.1038/s41586-020-2133-z>.

Correspondence and requests for materials should be addressed to G.H.

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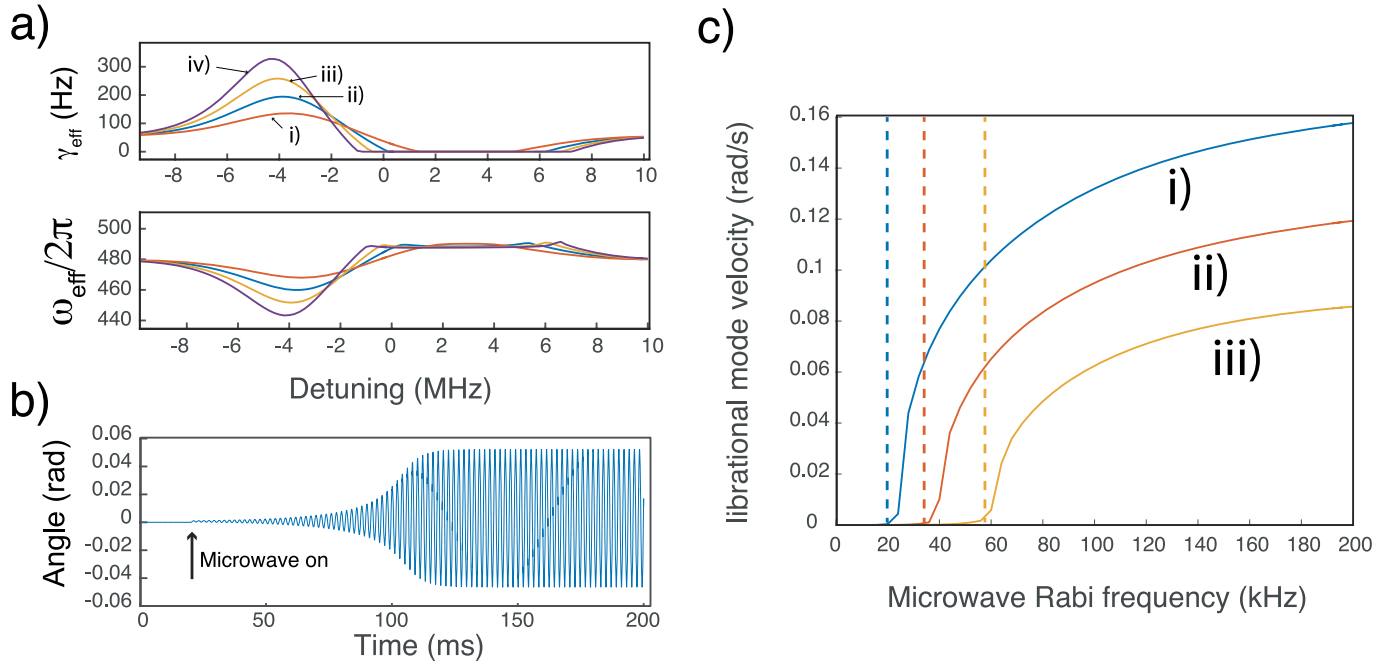
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Extended Data Fig. 1 | Parametric spin-excitation of the librational mode.

a, Detection principle of the librational modes using microwave parametric excitation (top). Reflected light field amplitude as a function of time for one low-frequency librational mode (170 Hz) at 1 mbar (bottom). **b,** Spin-mechanical resonance for two different microwave powers: trace i, 20 dBm; trace ii, -10 dBm. **c,** Reflected light field amplitude as a function of time, for the

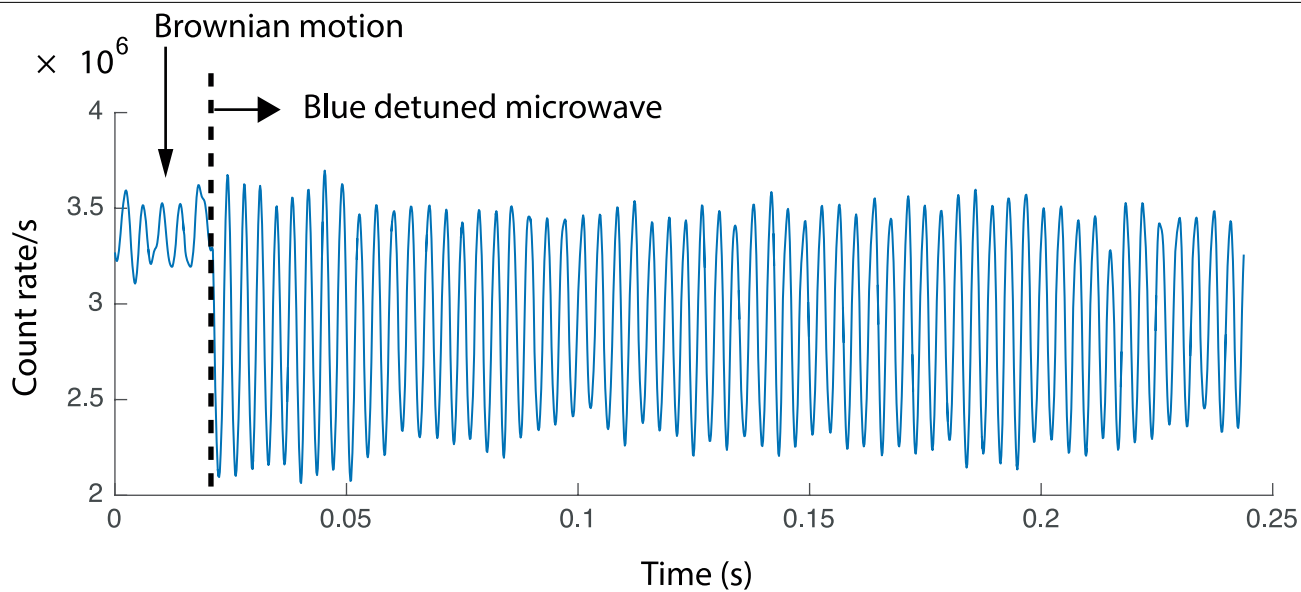
three microwave frequencies 2.617 GHz (red thin curve), 2.623 GHz (green) and 2.634 GHz (blue) corresponding to the points 1, 2 and 3 in **b**, respectively. Plain thicker lines show the slow envelope of the fit to the data (see Methods) using two sines at the frequency of the parametrically excited librational mode (480 Hz) and of the closest librational mode (590 Hz) with an amplitude that is 30 times smaller.



Extended Data Fig. 2 | Numerical simulations in the nonlinear regime.

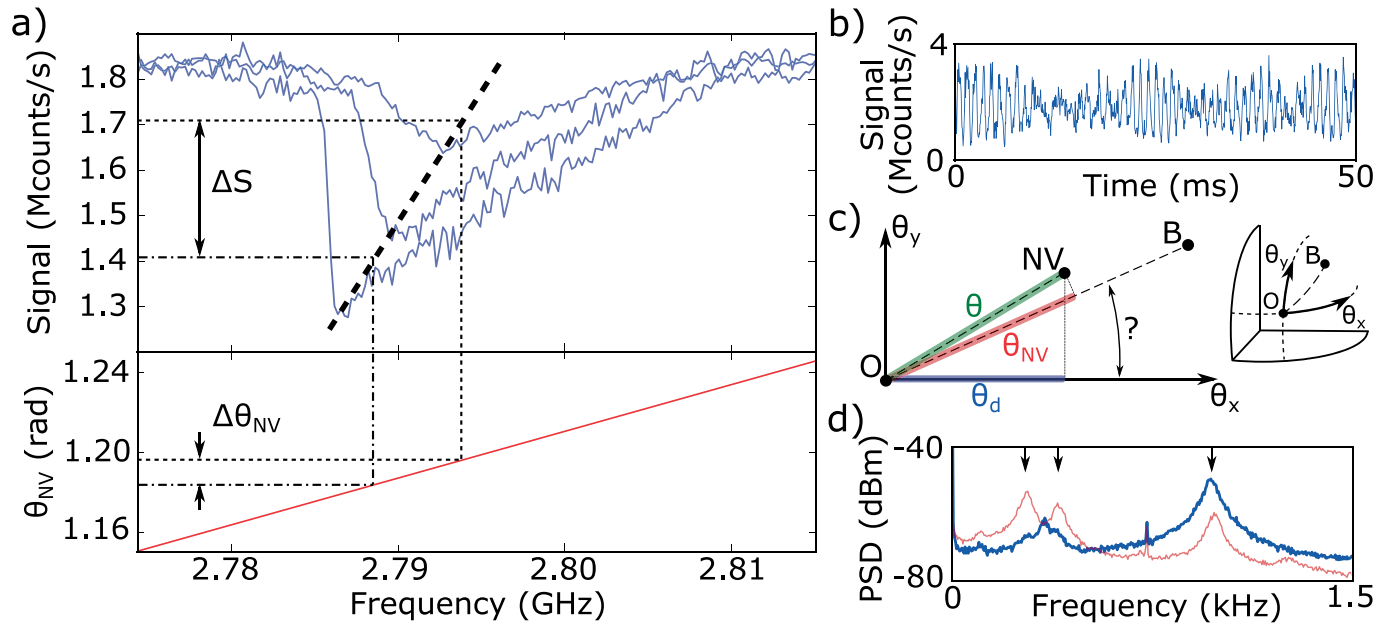
a, Effective damping and spring constants as a function of microwave detuning deduced by a numerical fit to the ring-down curves (see Methods). Traces i, ii, iii and iv correspond to increasing microwave Rabi frequencies $\Omega/2\pi = 50, 150, 200$ and 250 kHz, respectively. **b**, Evolution of the angle as a function of time

after turning on a microwave frequency to the blue of the spin resonance at a time $t = 20$ ms, showing amplification and finally lasing at $t = 120$ ms. **c**, Librational mode velocity as a function of microwave Rabi frequency, for three different values of the spin-lattice relaxation rate $1/T_1 = 1, 2$ and 3 kHz for traces i, ii and iii, respectively.



Extended Data Fig. 3 | Phonon-lasing as a function of time. Shown is the reflected light field amplitude as a function of time on sudden switch-on of a microwave signal at a time $t = 0.02$ s to the blue side of the spin transition.

The microwave power is above threshold so lasing can be observed after the Brownian motion signal (seen before $t = 0.02$ s). This curve was used to plot the histogram of Fig. 3C in the main text.

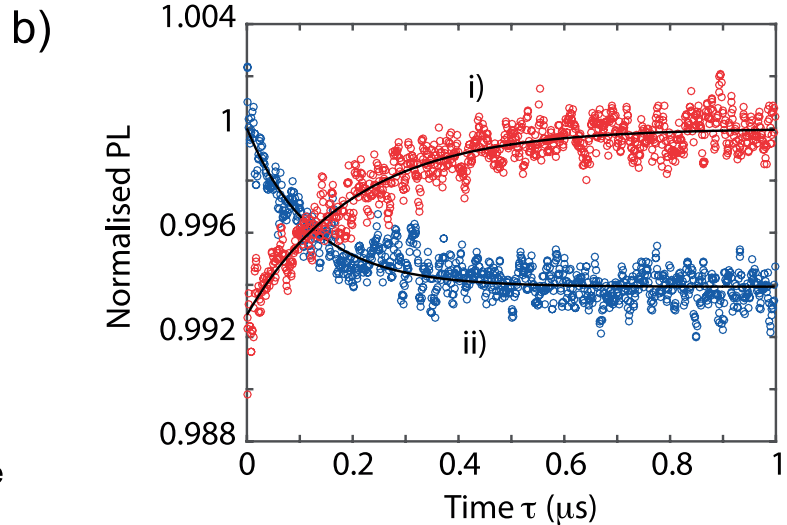
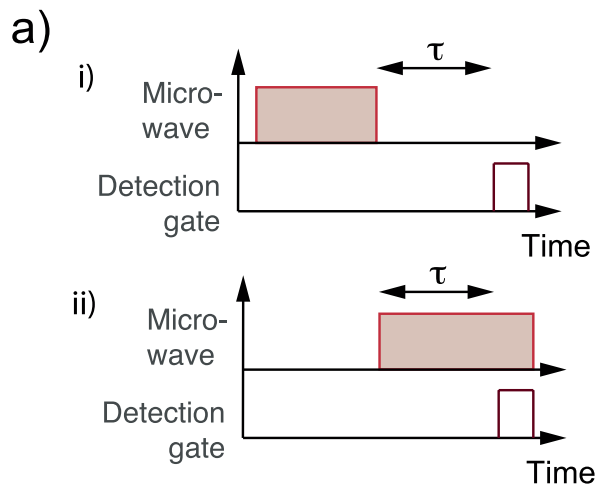


Extended Data Fig. 4 | Calibration of sensitivity using NV magnetometry.

a. Upper panel, mechanically detected spin resonance for three different microwave powers. The dashed line is the locus of the signal minima. Lower panel, angle between the NV axis and magnetic field direction versus NV spin transition frequency. ΔS is the maximal change of the optical signal. $\Delta\theta_{NV}$ is the maximum angle between the NV axis and the magnetic field direction.

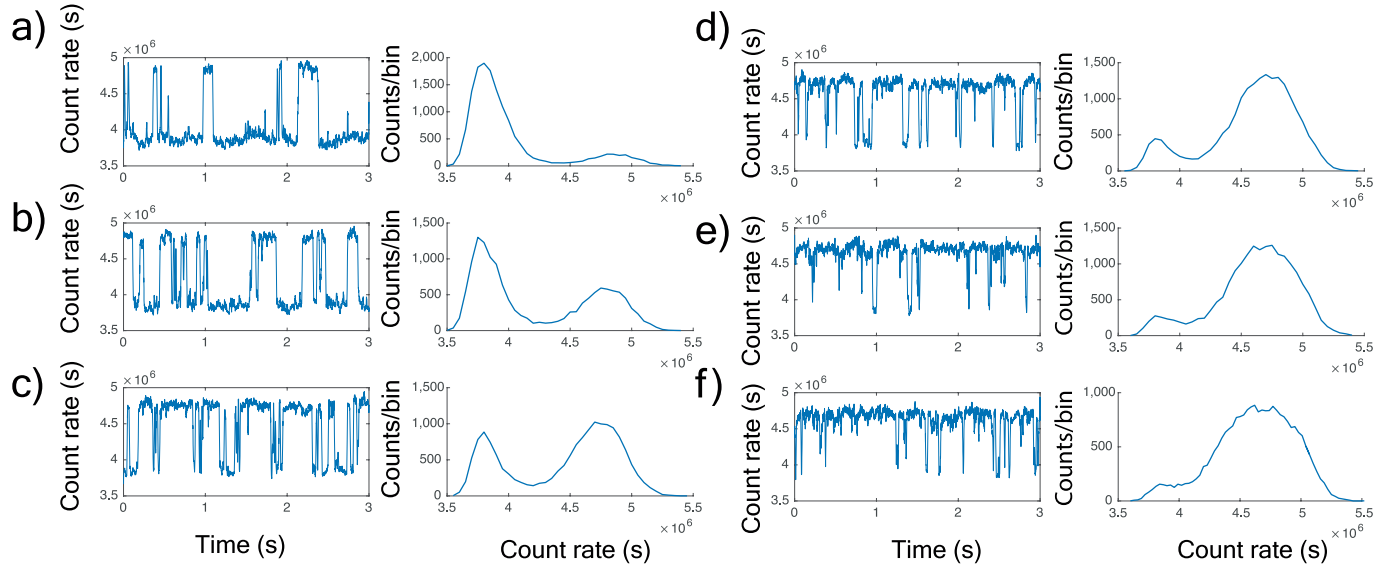
b. Optical signal as a function of time. **c.** Left panel, sketch depicting the angular motion of the diamond on magnetization of one class of NV spins using only two angles θ_x and θ_y for simplicity. B is the orientation of the magnetic

field. Right panel, angular trajectory represented in (θ_x, θ_y) space. We note O is the particle orientation without NV magnetization ($M_z = 0$). The red (green) line is the trajectory in the isotropic (anisotropic) case (see Methods). θ_d is the detected angle. **d.** Power spectra of the librational Brownian motion for two different speckle alignments taken with a resolution bandwidth of 1 Hz. The red curve shows all three librational modes. For the blue curve, the detection is tuned to be mainly sensitive to the mode with the highest confinement frequency. The latter detection setting is used for the data shown in **a** and **b**.



Extended Data Fig. 5 | Measurement of spin polarization rates. a, Sequences employed to measure the laser induced polarization rate to the ground state (i) and to measure the microwave induced polarization rate to the magnetic state $m'_s = -1$ (ii). The laser is kept on at all times for both sequences. b, Trace i shows the photoluminescence (PL) rate at a time τ after having turned off the

microwave signal. An exponential fit to the data gives a laser polarization time of 300 μs . Trace ii shows the photoluminescence rate at a time τ after having turned on the microwave signal. The polarization time to the magnetic state is here 124 μs .



Extended Data Fig. 6 | Particle angle dynamics for different microwave detunings in the bistable regime. a–f, Left panel, experimental observation of the reflected signal amplitude as a function of time for microwave frequencies

(in GHz) of 2.8000 (a), 2.8010 (b), 2.8020 (c), 2.8030 (d), 2.8035 (e), 2.8040 (f), tuned to the red of the spin transition. Right panel, histogram showing the number of counts within each bin.